

Lab Up For Success With Cisco Modeling Labs

Joe Clarke
Distinguished Engineer, Learn with Cisco



Lab Up For Success With Cisco Modeling Labs

Joe Clarke

PILOT







AI Generated



**Are you flaming out with your
certification exams?**

Practice smarter!!!

Agenda

1. Introduction to CML
2. Advanced CML Features
3. Making Friends with Exam Topics
4. Sample Labs

Session Goals

✓ **CML**
Intro to
intermediate use



✓ Combined with
Certification exam
prep techniques

Practice smarter
with CML

Get The Goods

Examples and code available on GitHub:

See this icon →



<https://github.com/CiscoLearning/ciscolive-brkcrt-2059>

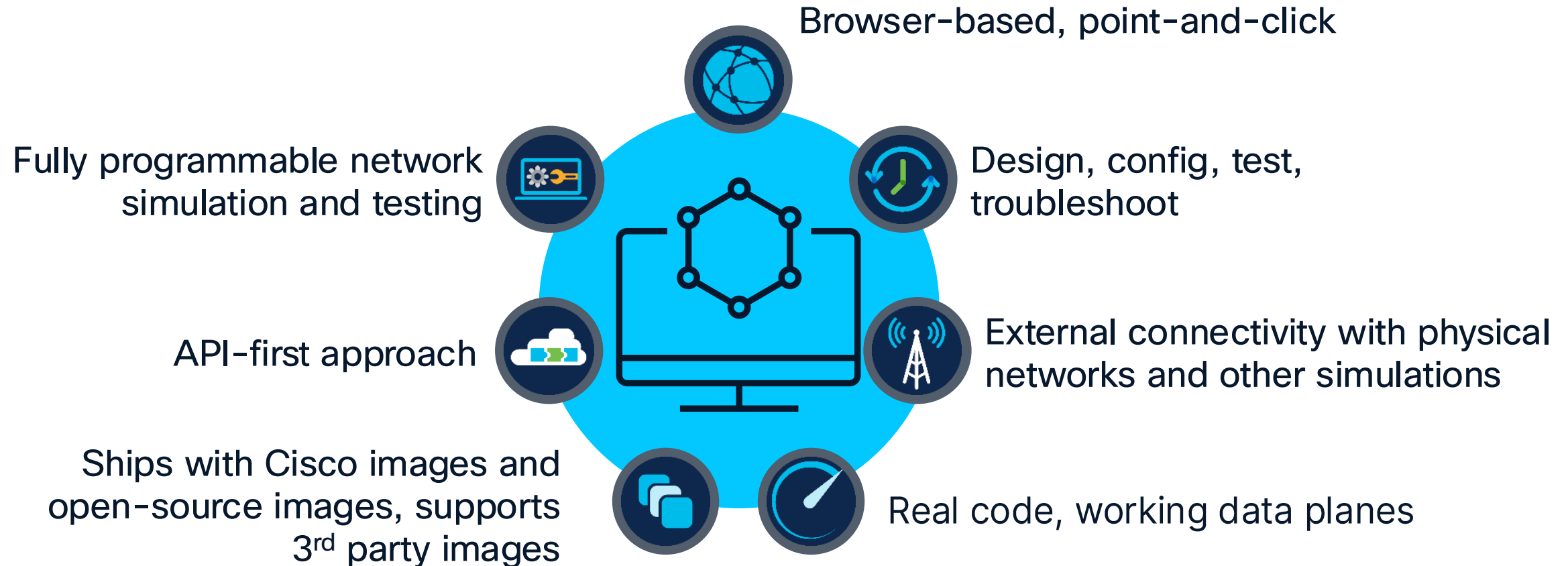




Introduction To CML

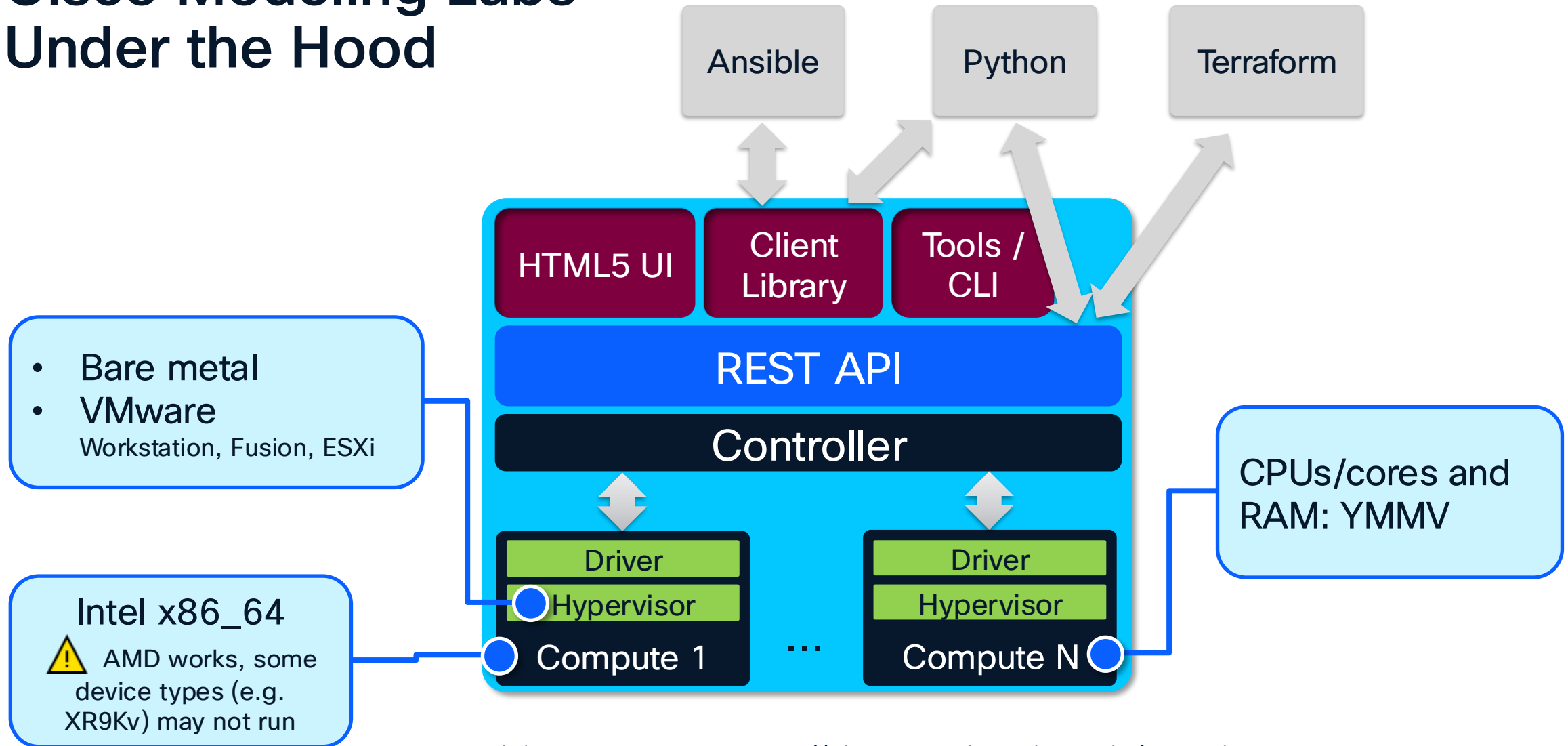
Meet Cisco Modeling Labs

Better than a Simulator!



Demo - The GUI

Cisco Modeling Labs – Under the Hood



Sizing Calculator: <https://ciscolearning.github.io/cml-sizer>

Flavors of CML

	Free	Personal	Personal Plus	Enterprise	Education
Multi-user					
Concurrent Nodes	5	20	40	520	
Clusterable					
Notes	Only IOL, IOL-L2, ASAv, and Linux nodes				Licensed based on number of students

CISCO Connect LatAm

But... but... I was told to get a hardware lab on ebay!

Hi! [Sign in](#) or [register](#)

Daily Deals

Brand Outlet

Gift Cards

Help & Contact

Sell

Watchlist ▾

My eBay ▾

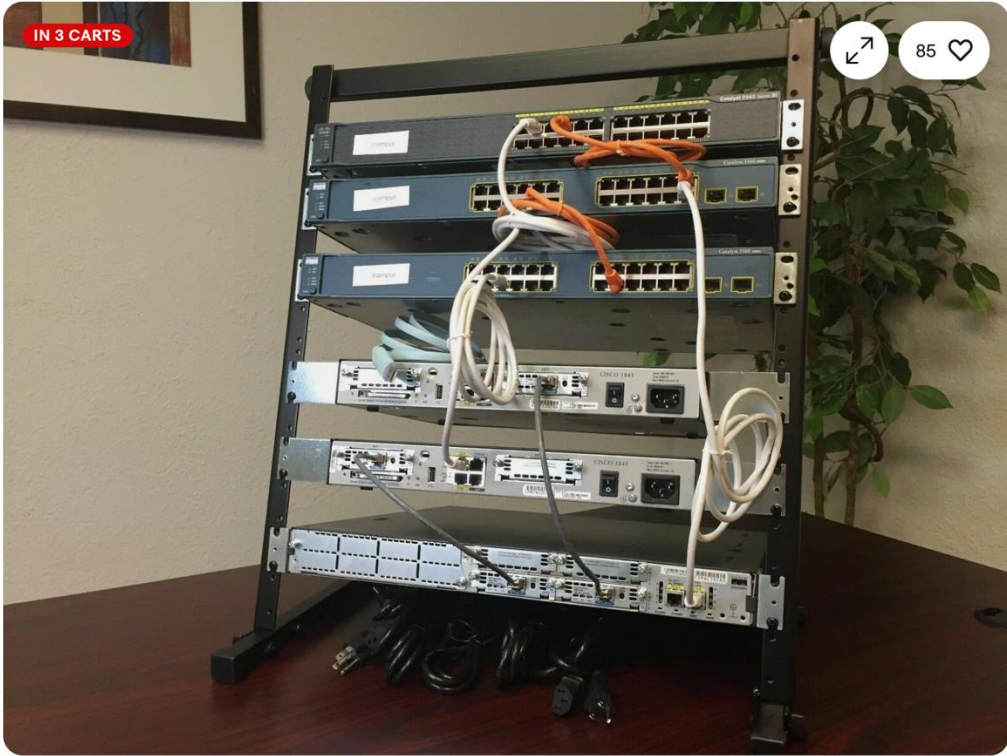
Shop by category ▾

All Categories ▾

Advanced

LIVE Streaming now Shop exclusive items from trusted sellers

IN 3 CARTS



85

NEW CISCO CCNA 200-301 CCNP v2.0 LAB KIT R&S

Hyper Network (4775)
100% positive · [Seller's other items](#) · [Contact seller](#)

US \$649.99 or Best Offer

as low as \$58.36/mo with **Klarna**. [Learn more](#)

Condition: **Used** ⓘ
"CISCO CCNA 200-301 CCNP v2.0 LAB KIT R&S
ROUTER IOS 15.7 & 2960 IOS 15.0 3560 IOS 12.1"

Quantity: **Last one · 84 sold**

© 2026 Cisco and/or its affiliates. All rights reserved.

BRKORT-2059

16

CISCO

So was he...





Practice, Practice, Practice

Always Be Labbing

Quickly build and destroy
virtual networks

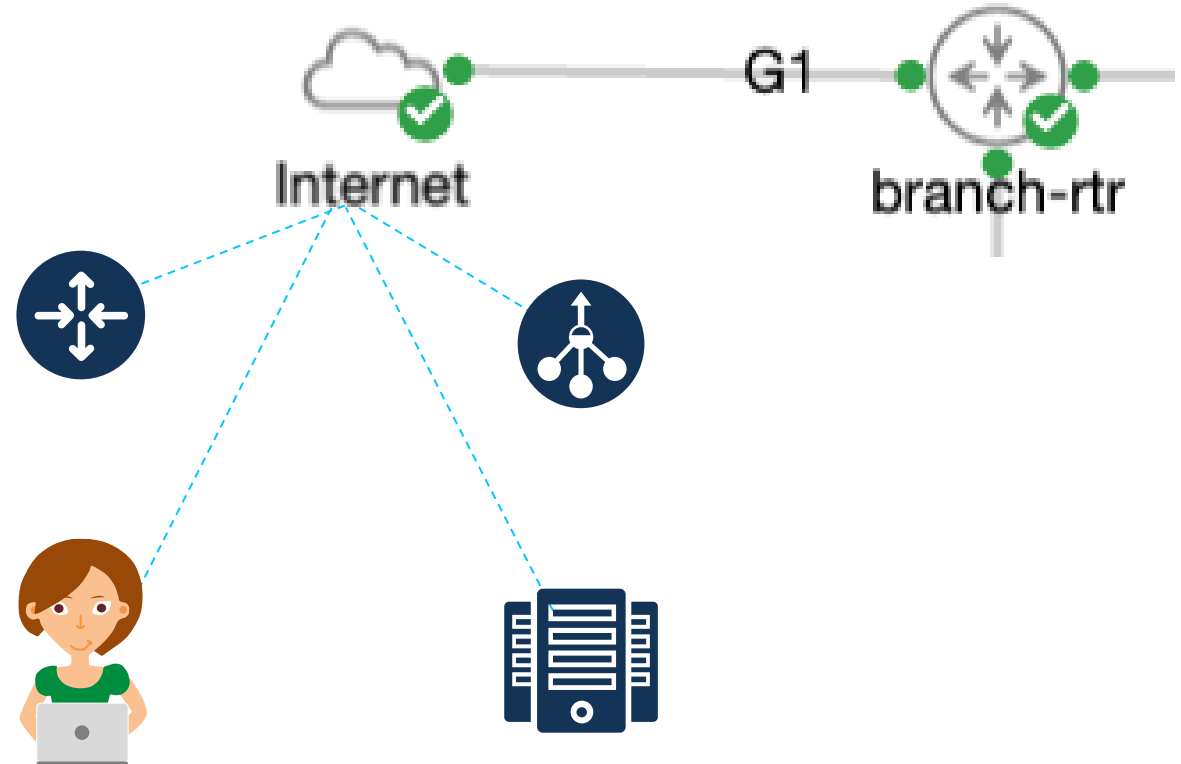
Share your network configurations

Advanced CML Features



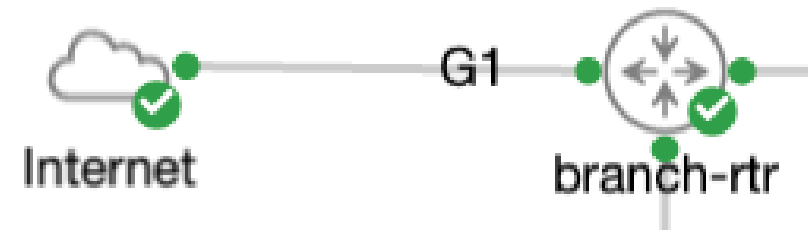
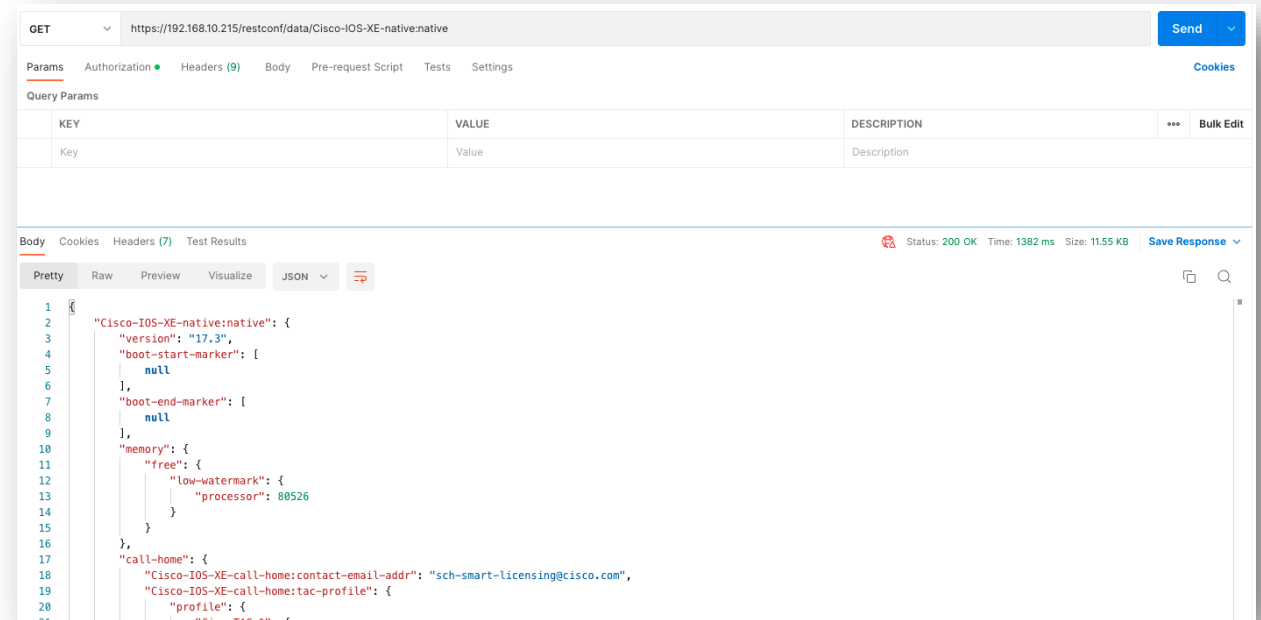
External Connectivity

- By default, CML labs are isolated
- External connectivity changes that
- Connect your labs to the world
 - NAT (L3) – think home networking
 - Bridge (L2) – share the segment on which the CML server sits
 - Custom (L2) – connect your labs to any layer 2 segment



External Connectivity Use Cases

- Download files
- Manage CML devices with external systems
- Access device APIs
- Augment your physical home lab



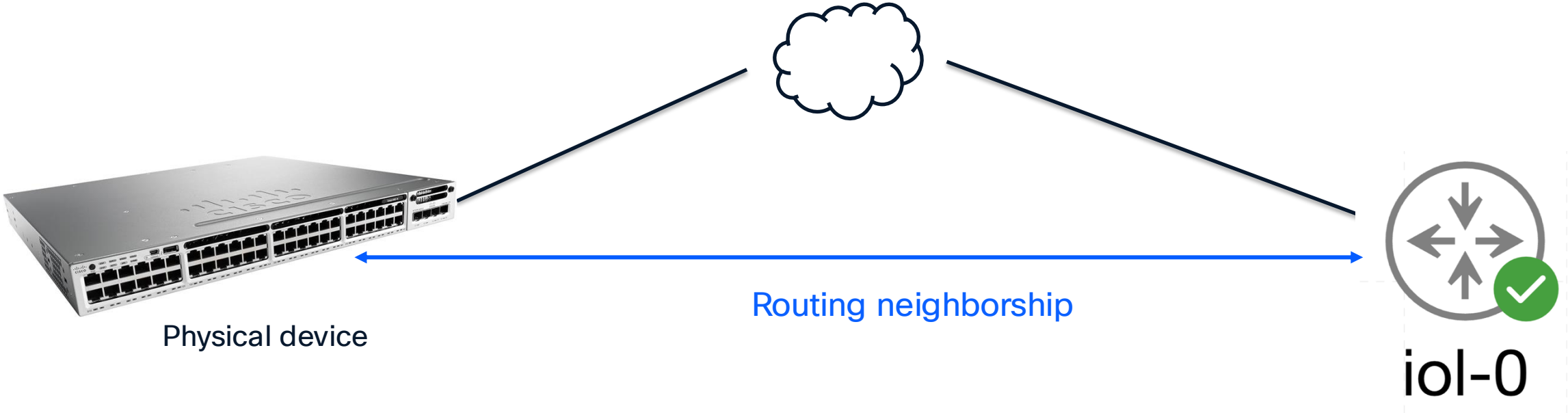
Augment Your Home Lab

Type of External Connectivity:

NAT

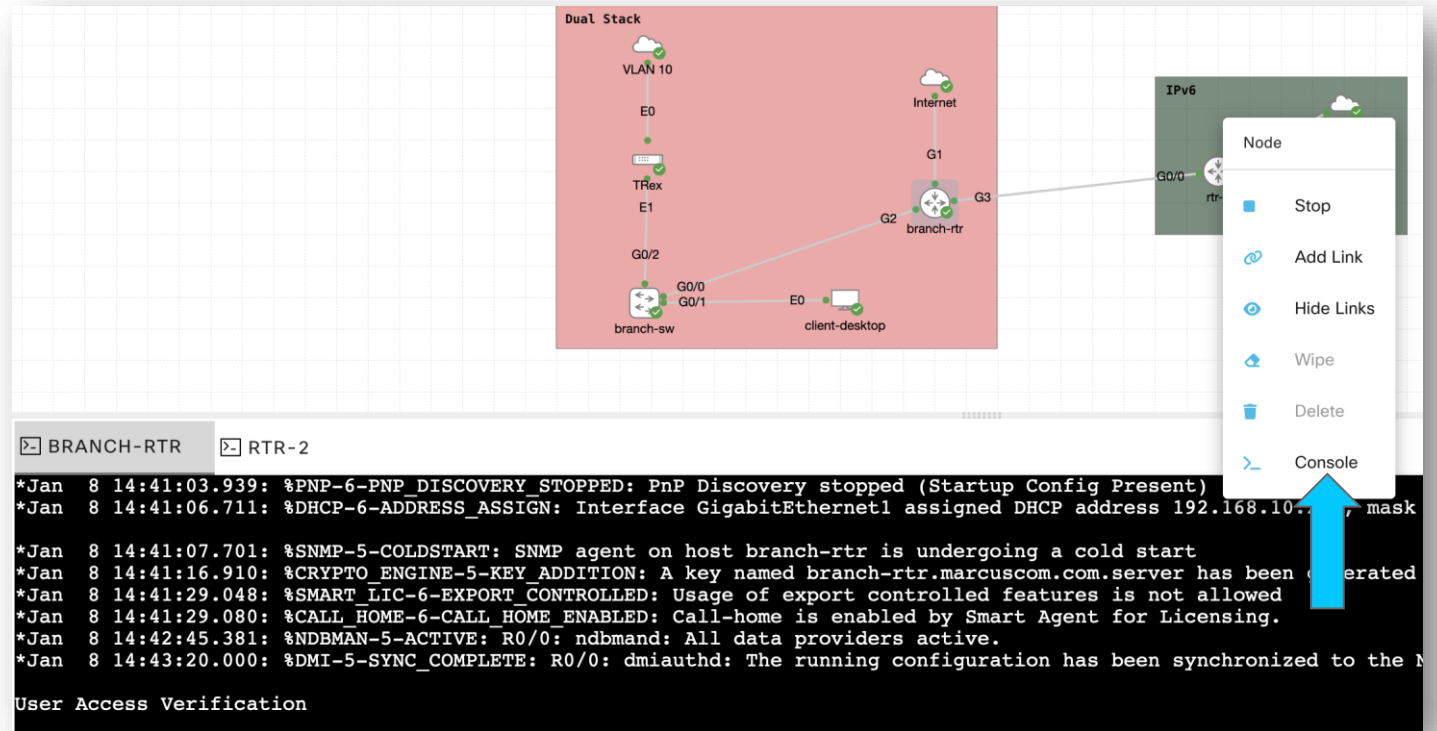
BRIDGE

CUSTOM



Terminal Console Pain

- The browser-based console is ok
- You might want to use your own terminal app
- Breakout Tool is cumbersome



PATty

- Enter **PATty**
- Console access (serial, VNC)
- Port Address Translation access



PATty – How to Configure Nodes

- Configuration is done using **tags** on **nodes**

- Available are

pat:outside:inside

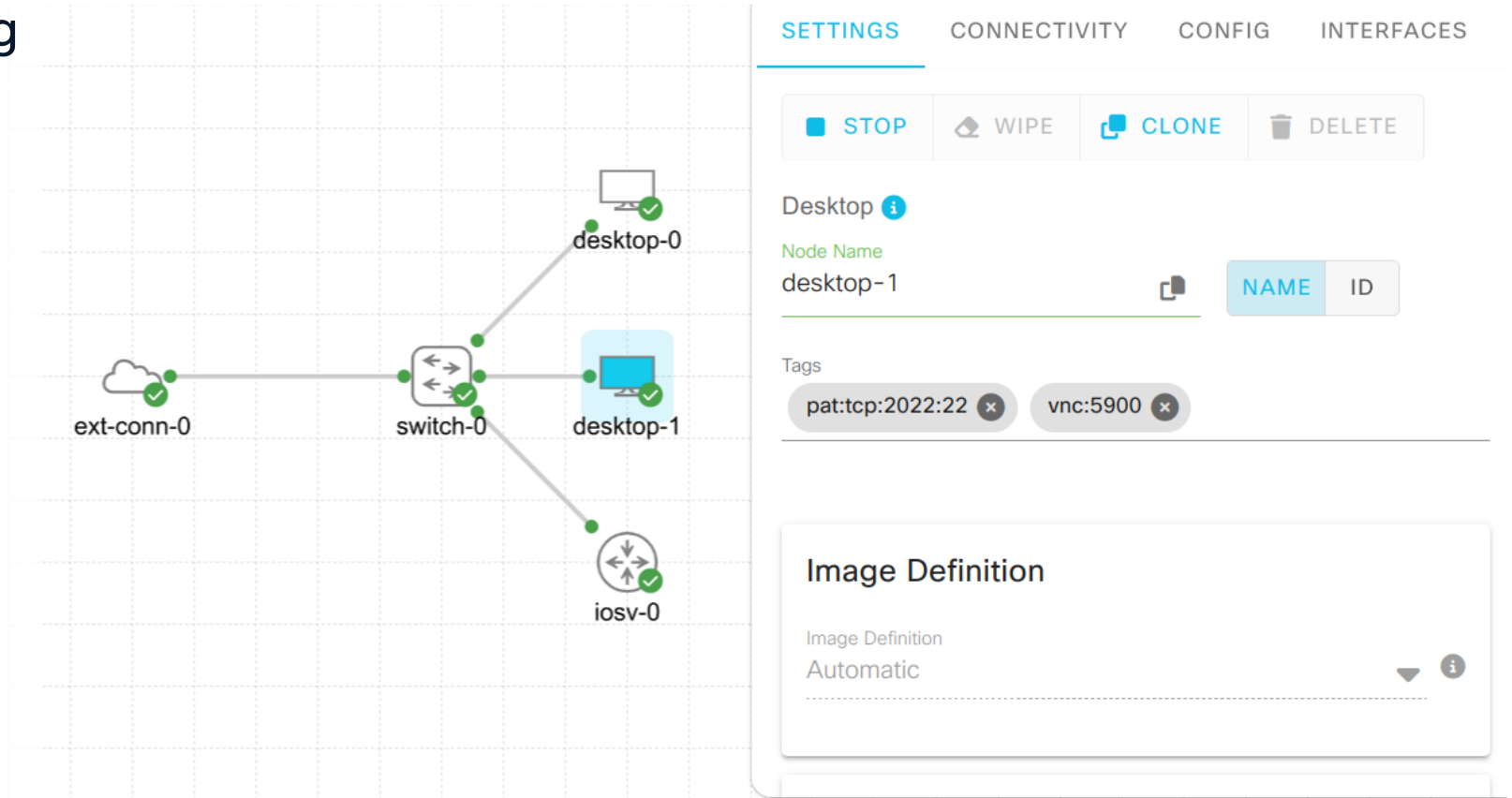
pat:tcp:outside:inside

pat:udp:outside:inside

serial:outside

serial:outside:linenum

vnc:outside



Using PATty Capabilities

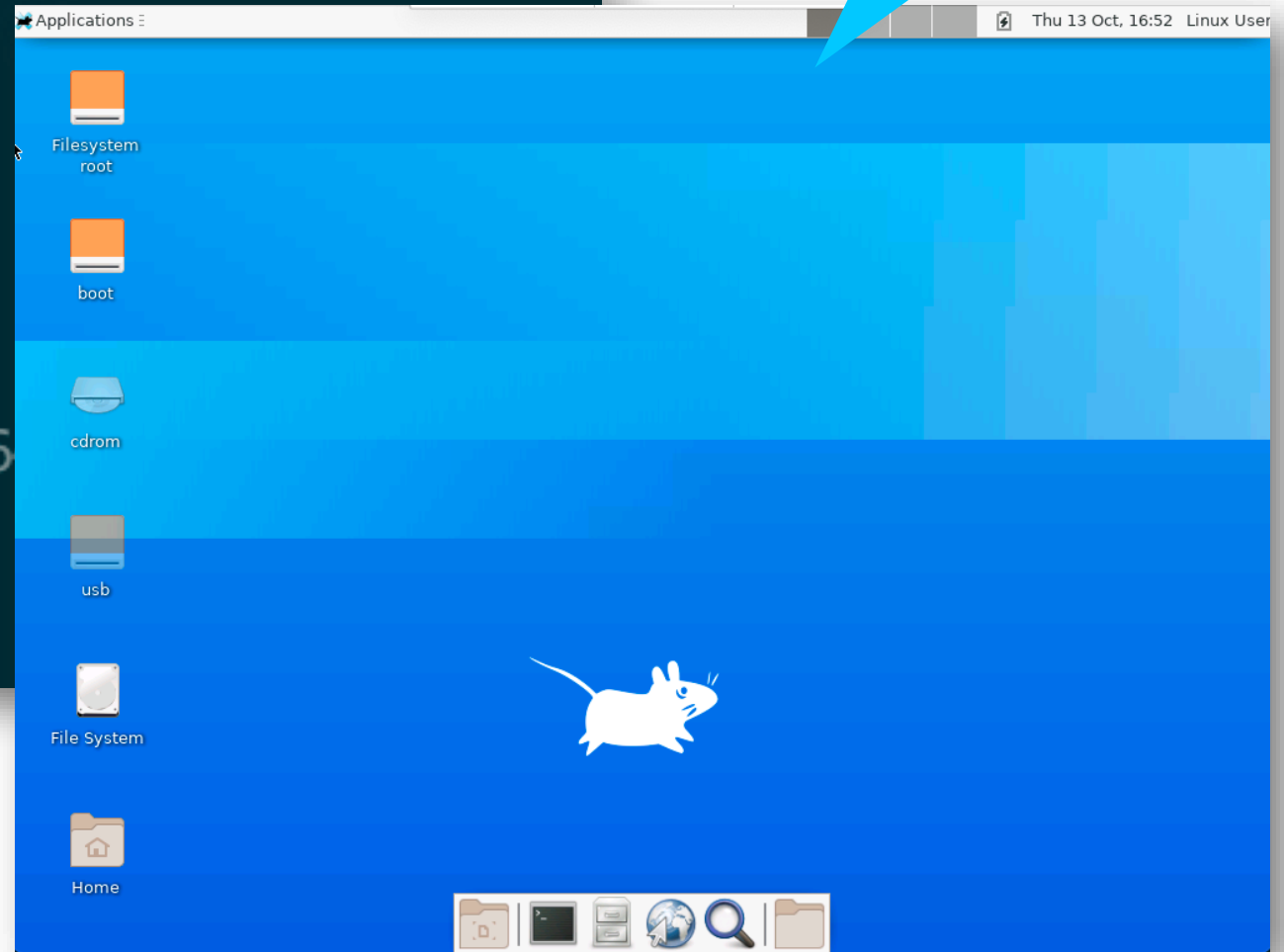
```
⇒ telnet localhost 9004
Trying ::1...
Connected to localhost.
Escape character is '^]'.

Welcome to Alpine Linux 3.12
Kernel 5.4.43-1-virt on an x86_64

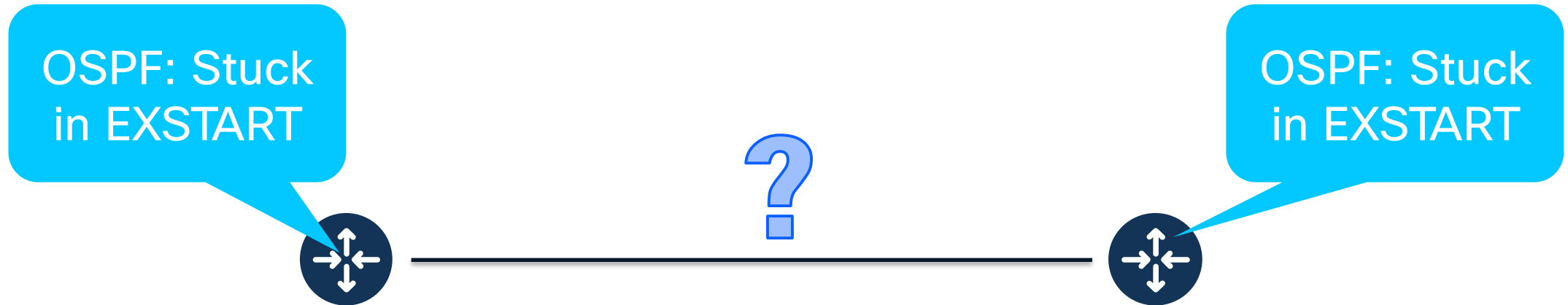
localhost login: █
```

serial:9004

vnc:5900

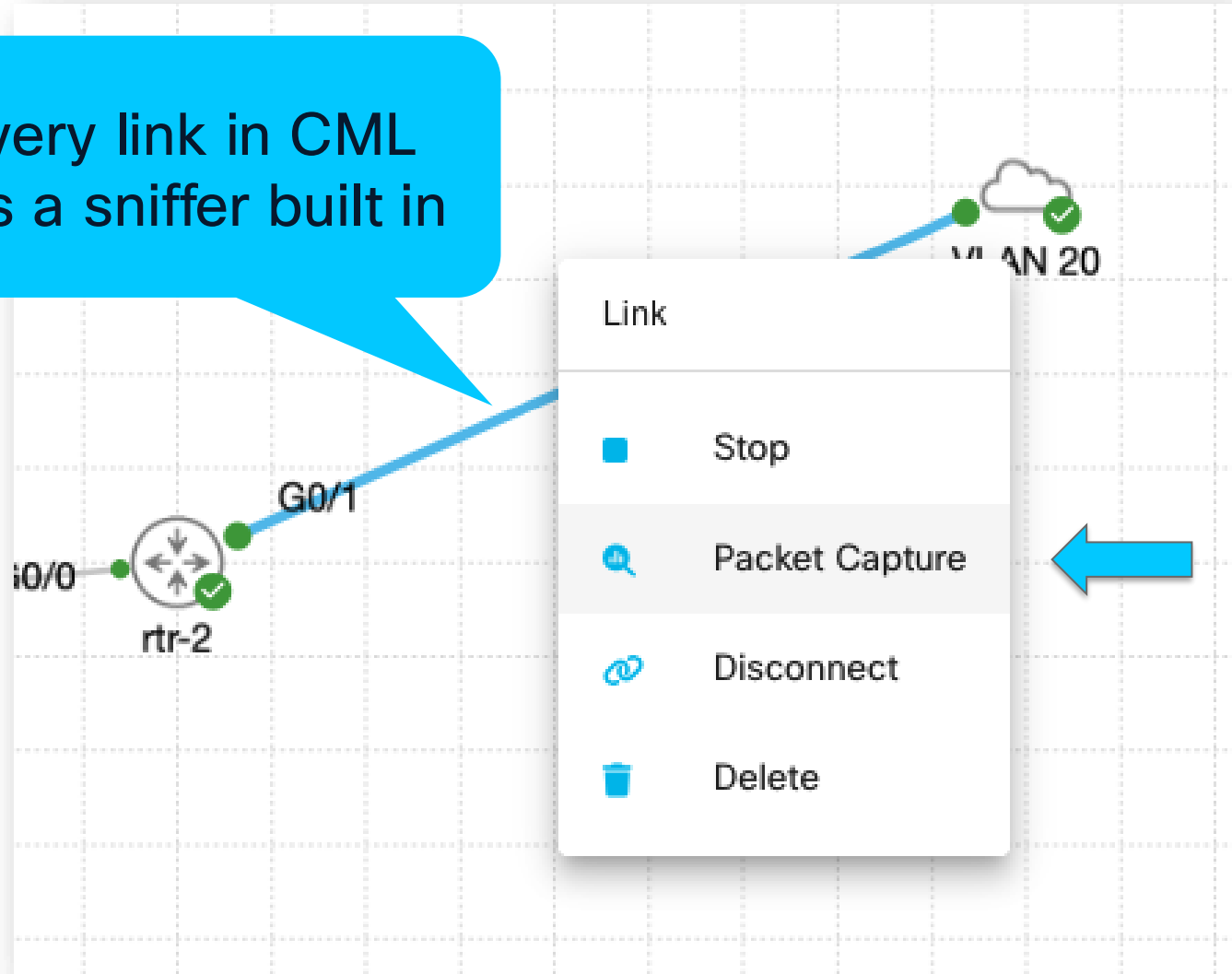


Fun With Links



Packet Captures Everywhere!

Every link in CML has a sniffer built in



Packet Capture Features

LINK INFO SIMULATE PACKET CAPTURE

▶ START 🗑️ CLEAR 📄 DOWNLOAD ⚙️ SETTINGS

🔍 Search

Max Packets:
50

Max Time:

BPF Templates:

DHCP

Spanning Tree

ip (src)

ip (dst)

ethernet (src)

ethernet (dst)

BPF:
udp port 68 or port 67



Search for packets

Download as a PCAP file

Berkeley Packet Filtering

Packet Capture Features

LINK INFOSIMULATEPACKET CAPTURE

▶ START

🗑️ CLEAR

⬇️ DOWNLOAD

⚙️ SETTINGS

🔍 Search

No.

Time

Source ▼

Destination ▼

Protocol ▼

Length ▼

Info ▼

Click start to begin capturing data

⚙️

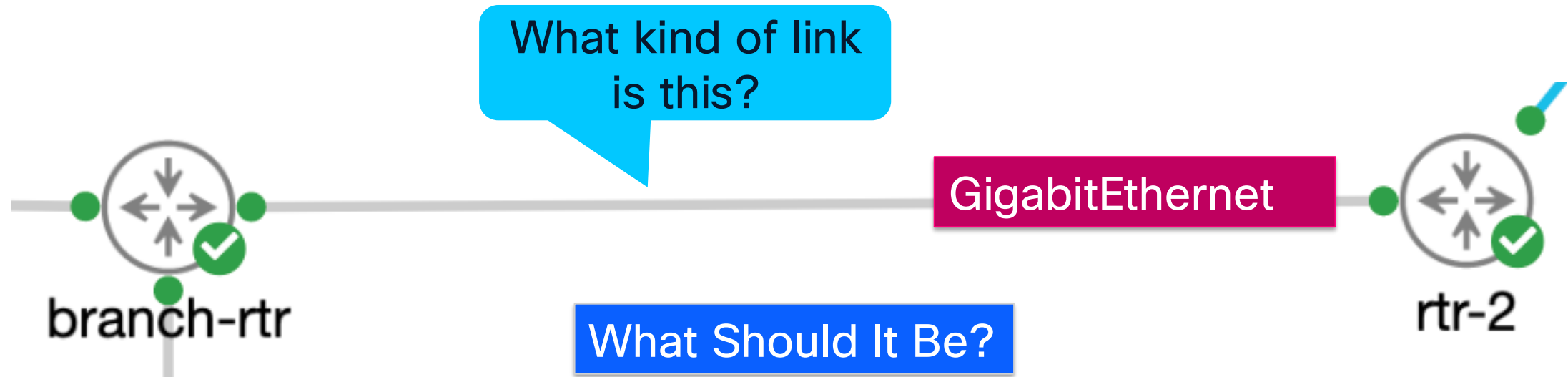
Packet Details

No packet selected

● Capture not running

Rows per page: 10 ▼ - < >

Not All Links Are Equal



Public Service Announcement: Dataplane Gotchas

- All links on CML are Gigabit Ethernet in *type*
 - N9Kv shows them as being 10 Gbps
- No interfaces named TenGE, HunGE, ...
 - Limitation of the underlying network elements
 - Cat9Kv allows you to change to TenGE
- Some devices (e.g. Linux nodes) can achieve ~ 1 Gbps speed
- Most have a rate-limited dataplane
 - See <https://developer.cisco.com/docs/modeling-labs/#!vm-images-for-cml-labs> for per-node dataplane limitations

Demo – Link Conditioning



What Are You Practicing Towards?



Making Friends With Exam Topics



Start with the Blueprint

350-401 ENCOR Exam Topics

Exam Description

To earn your CCNP Enterprise, CCIE Enterprise Infrastructure, or CCIE Enterprise Wireless certification, you must pass the **350-401 ENCOR** exam. This exam tests your knowledge of:


Architecture


Virtualization


Infrastructure


Network Assurance


Security

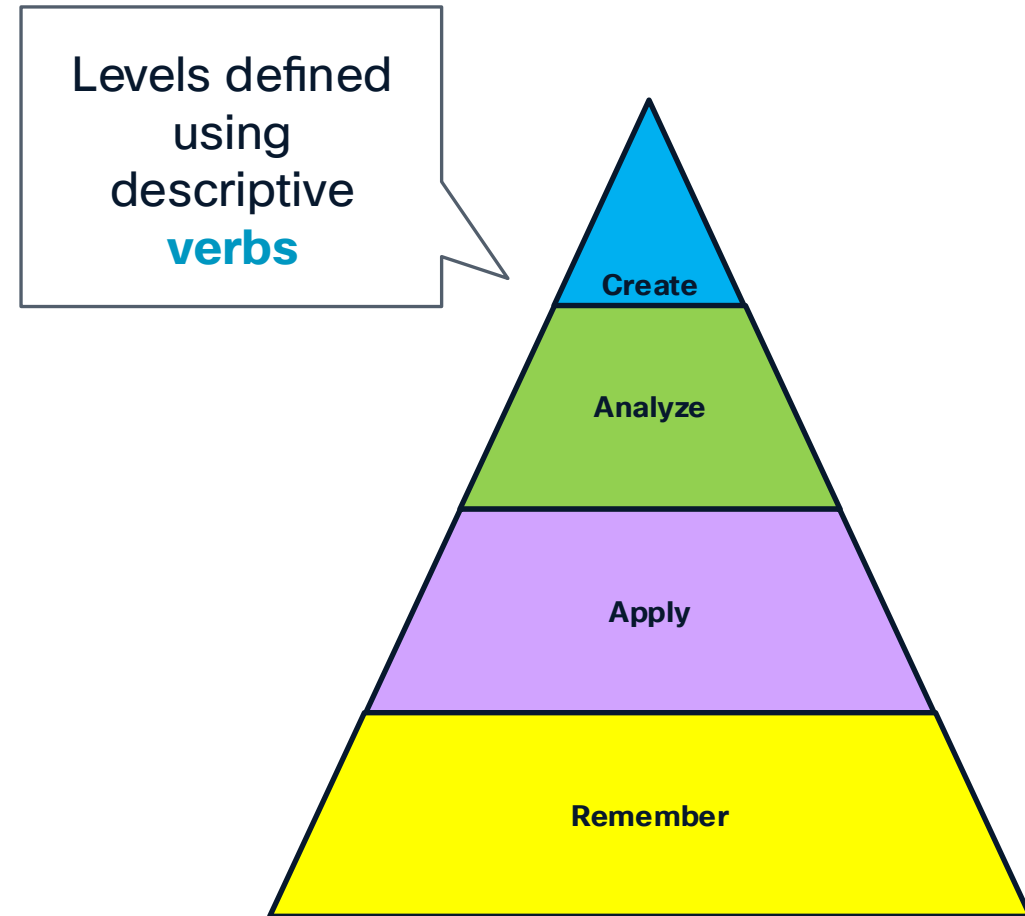

Automation

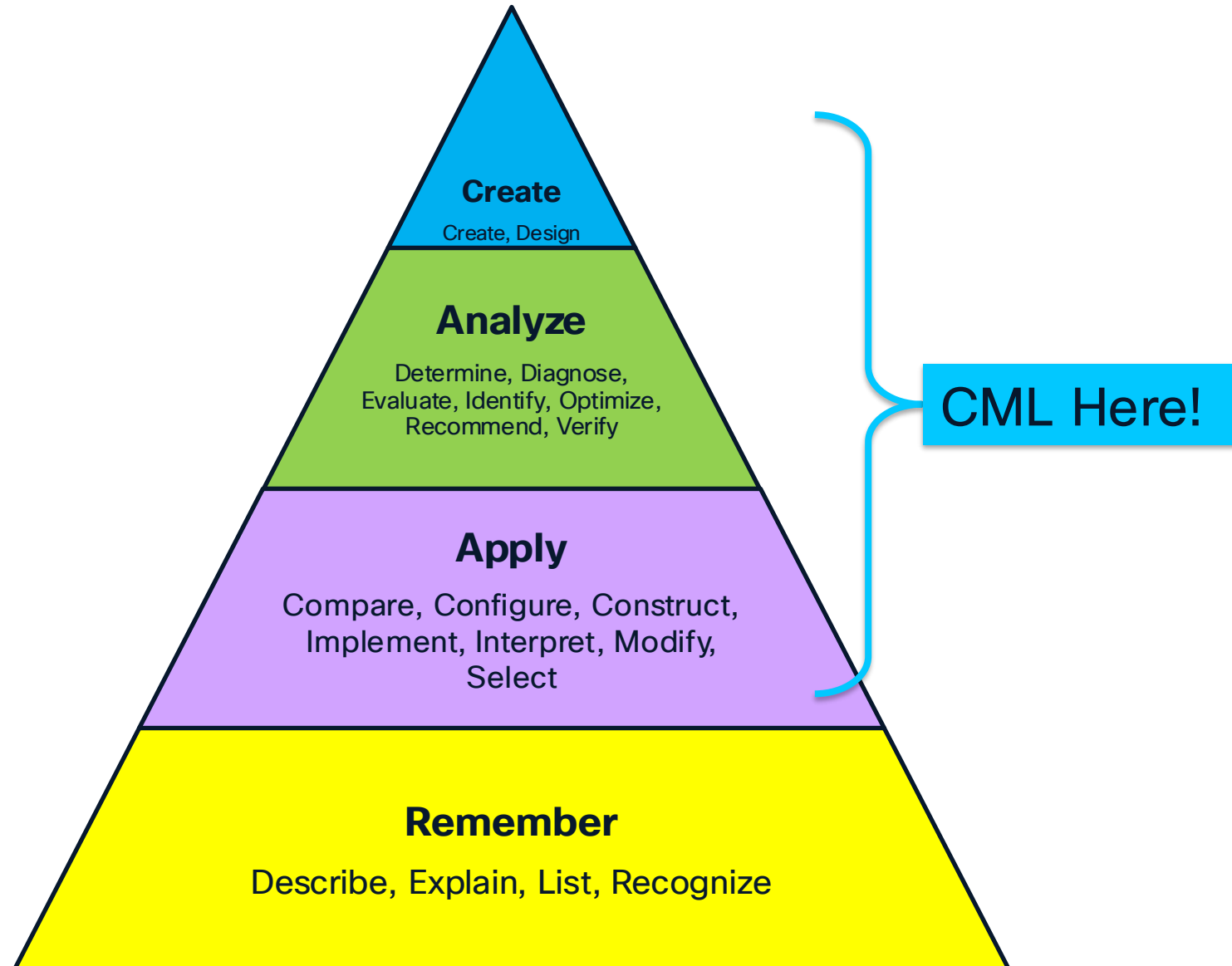
Expand each item below to view related exam topics.

1.0 Architecture	15%	▼
2.0 Virtualization	10%	▼
3.0 Infrastructure	30%	▼
4.0 Network Assurance	10%	▼
5.0 Security	20%	▼
6.0 Automation	15%	▼

How To Read Exam Topics:

Bloom's Taxonomy is used to convey the *cognitive complexity* of tasks





Consider ENCOR

350-401 ENCOR Exam Topics

Exam Description

To earn your CCNP Enterprise, CCIE Enterprise Infrastructure, or CCIE Enterprise Wireless certification, you must pass the **350-401 ENCOR** exam. This exam tests your knowledge of:



Architecture



Virtualization



Infrastructure



Network Assurance



Security



Automation

Expand each item below to view related exam topics.

1.0 Architecture	15%	▼
2.0 Virtualization	10%	▼
3.0 Infrastructure	30%	▼
4.0 Network Assurance	10%	▼
5.0 Security	20%	▼
6.0 Automation	15%	▼

3.1 Layer 2

- 3.1.a **Troubleshoot** static and dynamic 802.1q trunking protocols
- 3.1.b Troubleshoot static and dynamic EtherChannels
- 3.1.c Configure and verify common Spanning Tree Protocols (RSTP, MST) and Spanning Tree enhancements such as root guard and BPDU guard

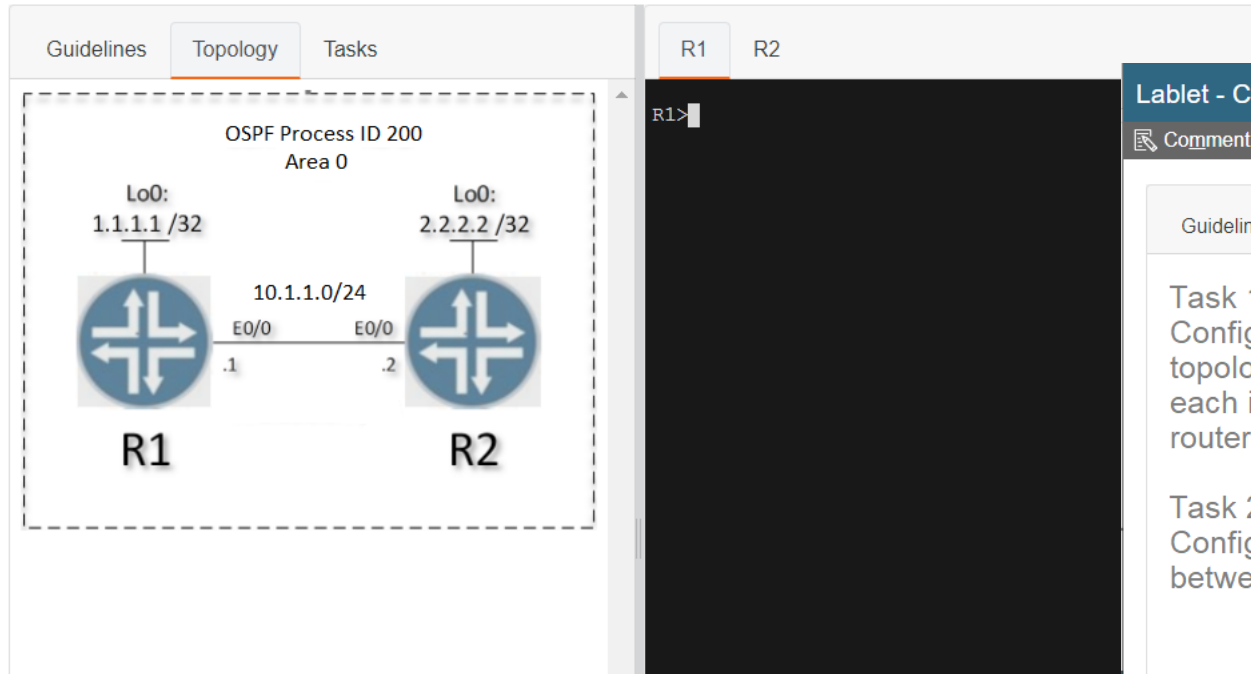
3.2 Layer 3

- 3.2.a Compare routing concepts of EIGRP and OSPF (advanced distance vector vs. link state, load balancing, path selection, path operations, metrics, and area types)
- 3.2.b **Configure** simple OSPFv2/v3 environments, including multiple normal areas, summarization, and filtering (neighbor adjacency, point-to-point, and broadcast network types, and passive-interface)
- 3.2.c Configure and **verify** eBGP between directly connected neighbors (best path selection algorithm and neighbor relationships)
- 3.2.d Describe policy-based routing

Get Ready for Performance-based Lablets

Lablet - Candidate Name ⌚ Time Remaining 01:26:03 🔍 1 of 1

Comment



Lablet - Candidate Name ⌚ Time Remaining 01:17:16 🔍 1 of 1

Comment

Guidelines **Topology** **Tasks**

R1 R2

Task 1:
Configure OSPF according to the topology. Match the network mask of each interface and ensure that all routers use Lo0 for the router ID.

Task 2:
Configure OSPF MD5 authentication between R1 and R2.

[Next](#) ➔

Sample Labs

The background is a light gray gradient. A white rectangle is tilted diagonally from the top right towards the center. Inside this rectangle is a smooth color gradient that transitions from red on the left to yellow in the middle and then to blue on the right. In the bottom left corner, there is a large, light blue arc that curves upwards and to the right.

So Many Sample Labs!



CCNA Topology Contest submissions

- Entry #1 - 2.1 Configure and verify 1.1
- Entry #2 - 2.2 Configure and verify 1.2
- Entry #3 - 2.1 Configure and verify 1.3
- Entry #4 - 2.2 Configure and verify 1.4
- Entry #5 - 2.3 Configure and verify 1.5
- Entry #6 - 2.4 Configure and verify 1.6
- Entry #7 - 2.5 Configure and verify 1.7
- Entry #8 - 2.6 Configure and verify 1.8
- Entry #9 - 2.7 Configure and verify 1.9
- Entry #10 - 2.8 Configure and verify 1.10
- Entry #11 - 2.9 Configure and verify 1.11
- Entry #12 - 2.10 Configure and verify 1.12
- Entry #13 - 2.11 Configure and verify 1.13
- Entry #14 - 2.12 Configure and verify 1.14
- Entry #15 - 2.13 Configure and verify 1.15
- Entry #16 - 2.14 Configure and verify 1.16
- Entry #17 - 2.15 Configure and verify 1.17
- Entry #18 - 2.16 Configure and verify 1.18
- Entry #19 - 2.17 Configure and verify 1.19

CCNA Exam Prep: Back to Networking Basics with Hank Preston and Patrick Gargano

Mastering VLAN Configuration

Exploring why VLANs exist in network engineering and how to configure them on Cisco IOS switches.

- Creating VLANs
- Configuring Access Ports
- Configuring 802.1q trunk interfaces
- Inter VLAN Routing options

Setup and Scenario

In this set of lab-based demonstrations, you are the network engineer for a growing organization tasked with updating the network to support new network needs. The network traditionally has had 2 small network segments. One for "Employees" and a second one for "Guests".

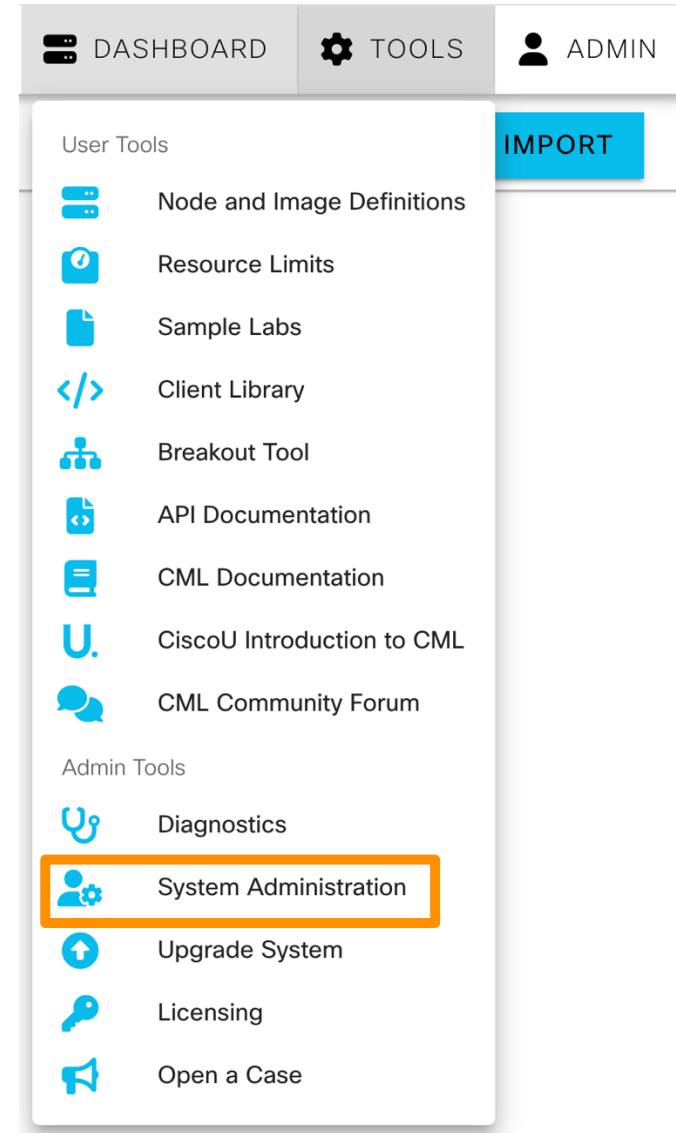
You are getting two requests from the business:

1. The facilities team is asking for network connectivity for IOT Devices. These devices need to be segmented from employee and guest traffic.
2. The organization is growing, and there is need to add network capacity to support additional users. These new users will be located in new office space being prepared in a different part of the building.

1. The facilities team is asking for network connectivity for IOT Devices. These devices need to be segmented from employee and guest traffic.

2. The organization is growing, and there is need to add network capacity to support additional users. These new users will be located in new office space being prepared in a different part of the building.

Load Them Straight From GitHub



Add the Repositories

Cisco Modeling Labs

Lab Repositories

DASHBOARD

TOOLS

ADMIN

SYSTEM MAINTENANCE

NOTICES

COMPUTE HOSTS

USER AUTHENTICATION

USER ADMINISTRATION

GROUP ADMINISTRATION

RESOURCE POOLS

NODE ADMINISTRATION

TELEMETRY

LAB REPOSITORIES

EXTERNAL CONNECTORS

HOME / LAB REPOSITORIES

Q Search

+ ADD

REFRESH ALL

Name	URL	Folder	Actions
cml-community	https://github.com/CiscoDevNet/cml-community	lab-topologies	REMOVE
ciscolive-brkcrt-2059	https://github.com/CiscoLearning/ciscolive-brkcrt-2059	sample-labs	REMOVE

Diagram illustrating a network configuration for Sample Lab 1 (VLAN Configuration). The diagram shows two VLANs, VLAN 10 and VLAN 20, connected via a central horizontal line. VLAN 10 is represented by a teal box at the top, and VLAN 20 is represented by a tan box at the bottom. Each VLAN has two ports. A horizontal line connects the two ports of VLAN 20, and a vertical line connects the two ports of VLAN 10. A central horizontal line connects the two vertical lines, representing a switch or router interface.

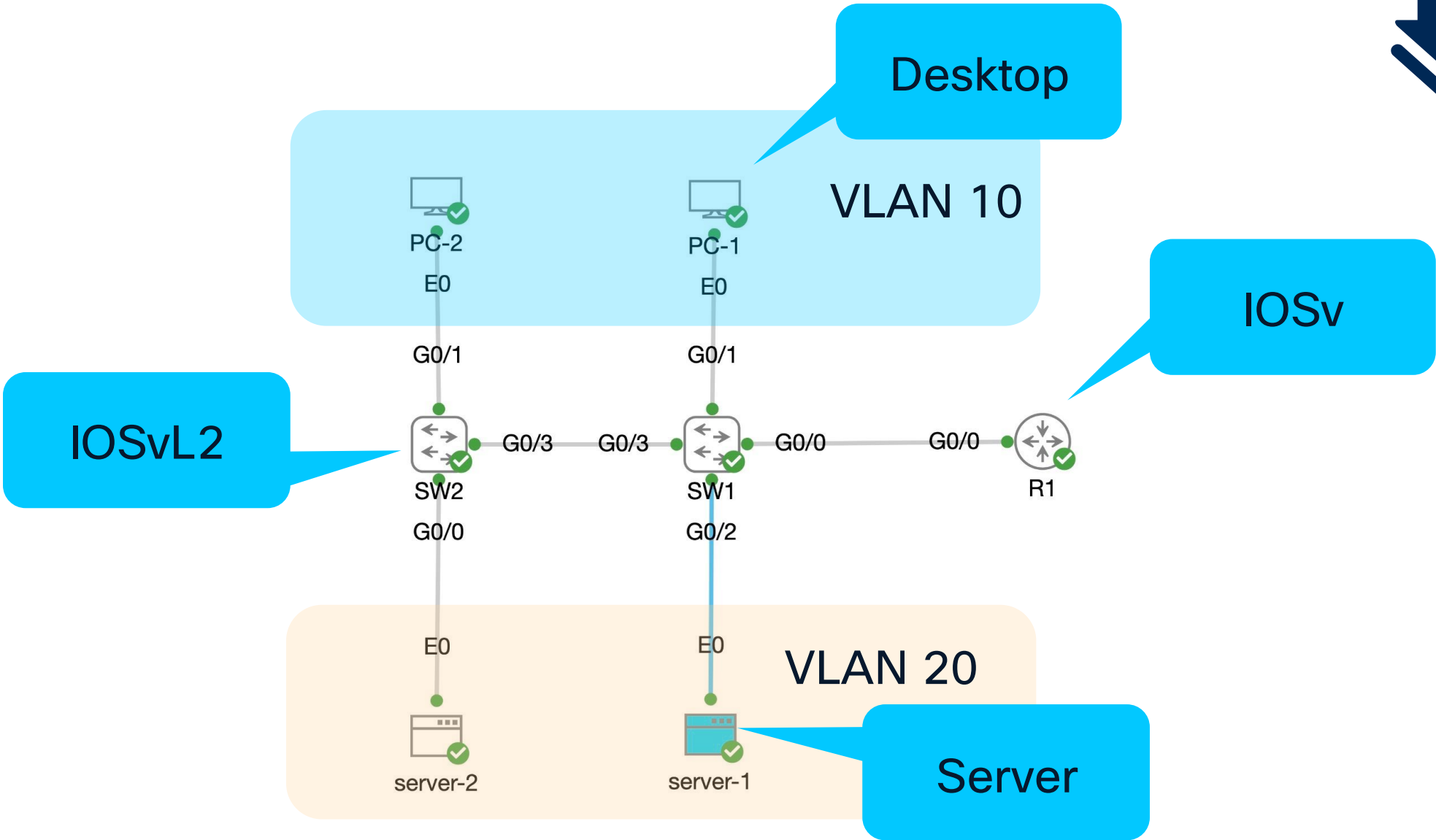
Sample Lab 1: CCNA

Implement Multiple VLANs and Basic Routing Between the VLANs

- Implement VLAN assignments on two switches
- Configure a dot1q trunk between those switches
- Configure a dot1q trunk between one switch and a router that will handle the routing for the VLANs
- Configure VLAN trunking on that router to act as a gateway between VLANs
- Verify that connectivity exists between hosts in the lab on each VLAN, and that the connectivity goes through the router

Sample Lab 1: CCNA

Topology



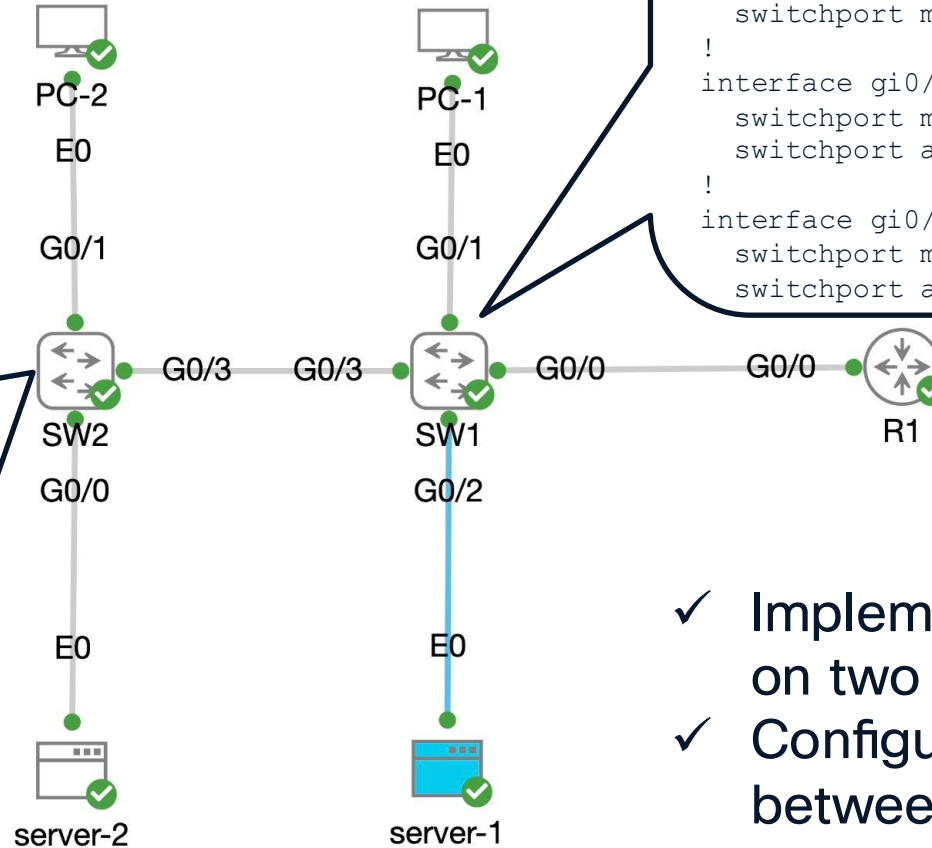
Sample Lab 1: CCNA

Config

```

vlan 10
  name Users
!
vlan 20
  name Servers
!
interface gi0/0
  switchport mode access
  switchport access vlan 20
!
interface gi0/1
  switchport mode access
  switchport access vlan 10
!
interface gi0/3
  switchport trunk encap dot1q
  switchport mode trunk

```



```

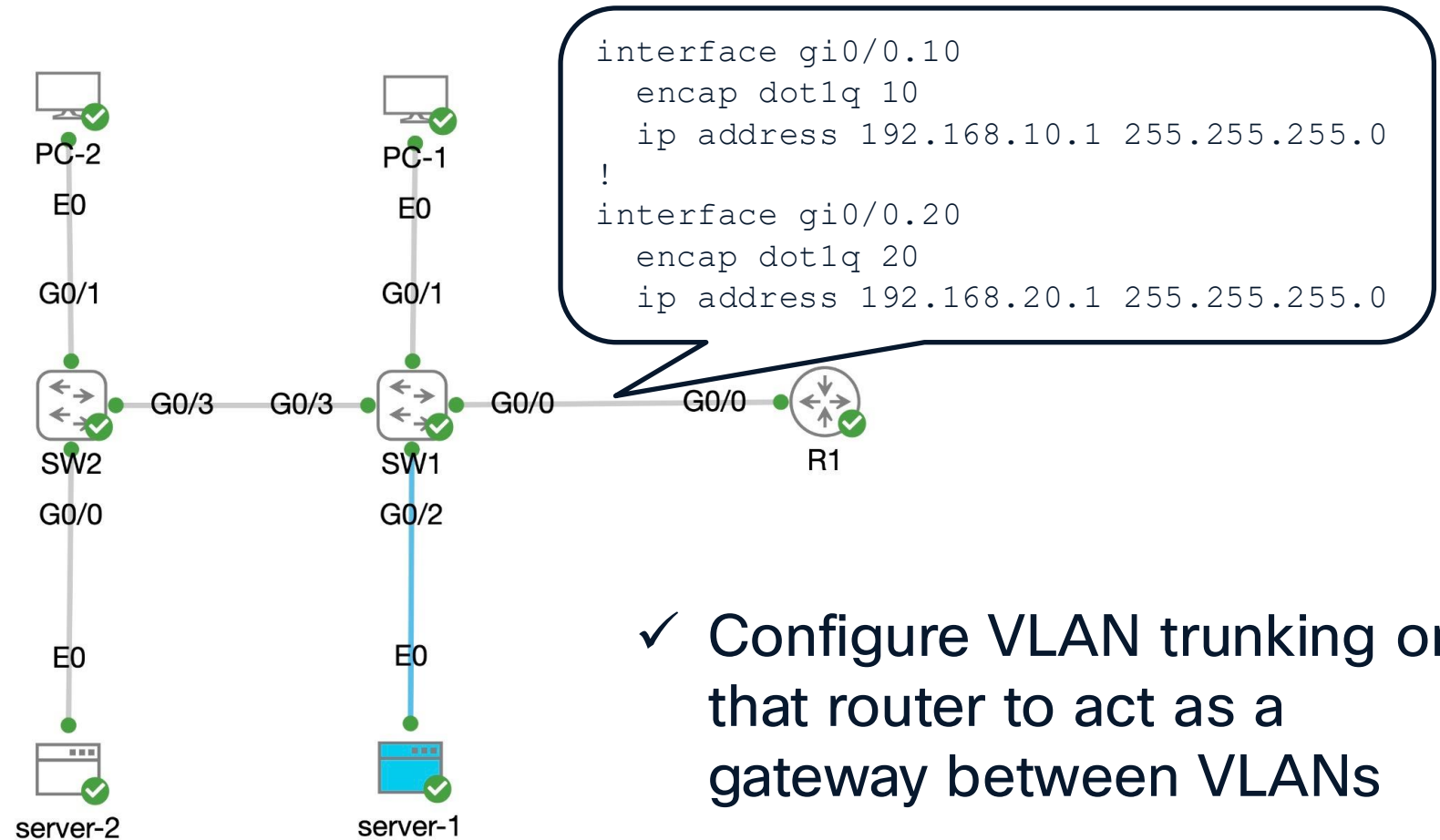
vlan 10
  name Users
!
vlan 20
  name Servers
!
interface gi0/0
  switchport trunk encap dot1q
  switchport mode trunk
!
interface gi0/1
  switchport mode access
  switchport access vlan 10
!
interface gi0/2
  switchport mode access
  switchport access vlan 20

```

- ✓ Implement VLAN assignments on two switches
- ✓ Configure a dot1q trunk between those switches

Sample Lab 1: CCNA

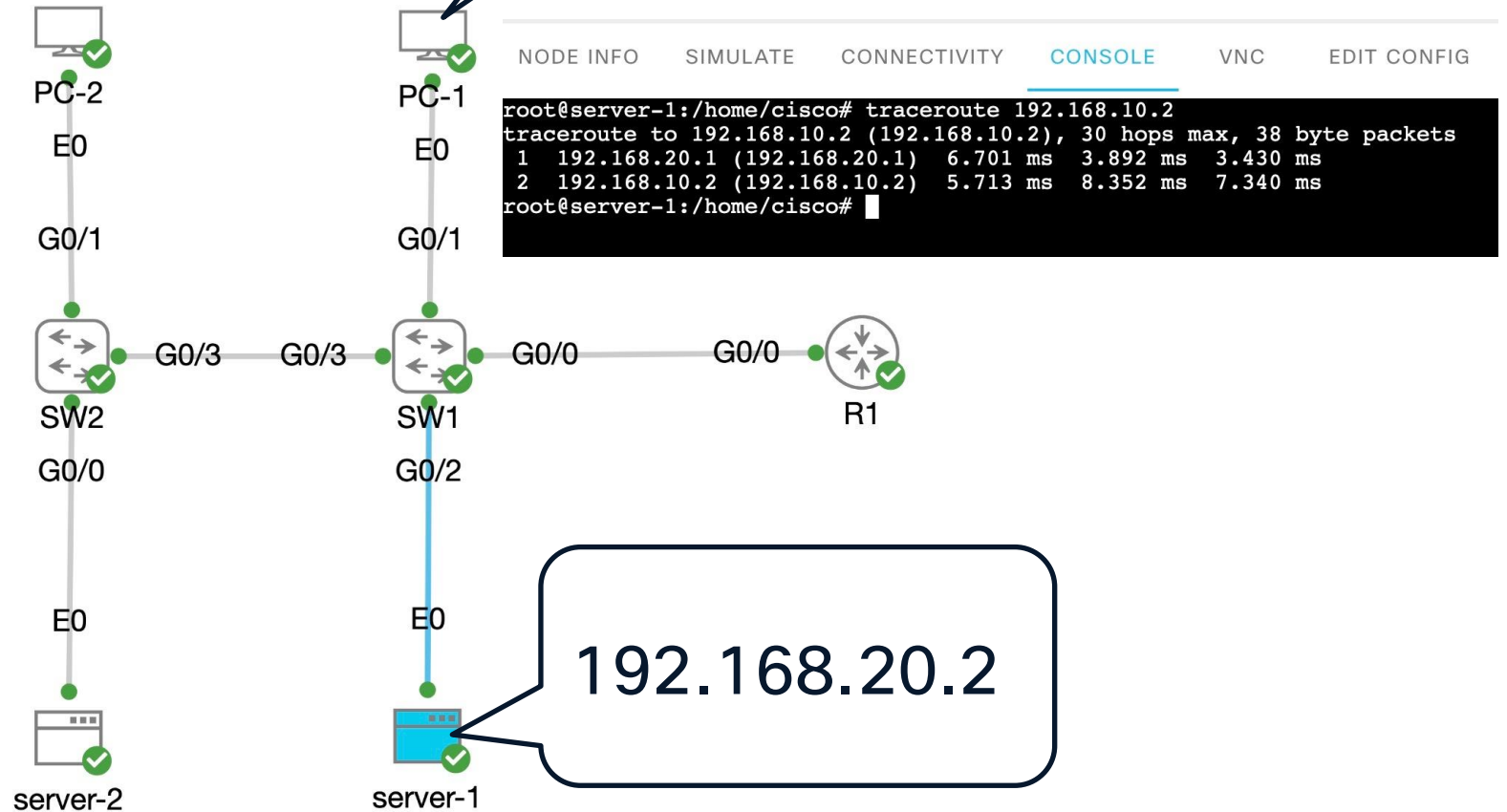
Config



Sample Lab 1: CCNA

Verify

- ✓ Verify that connectivity exists between hosts in the lab on each VLAN, and that the connectivity goes through the router



Sample Lab 2: Enterprise Advanced Routing and Switching (ENARSI)

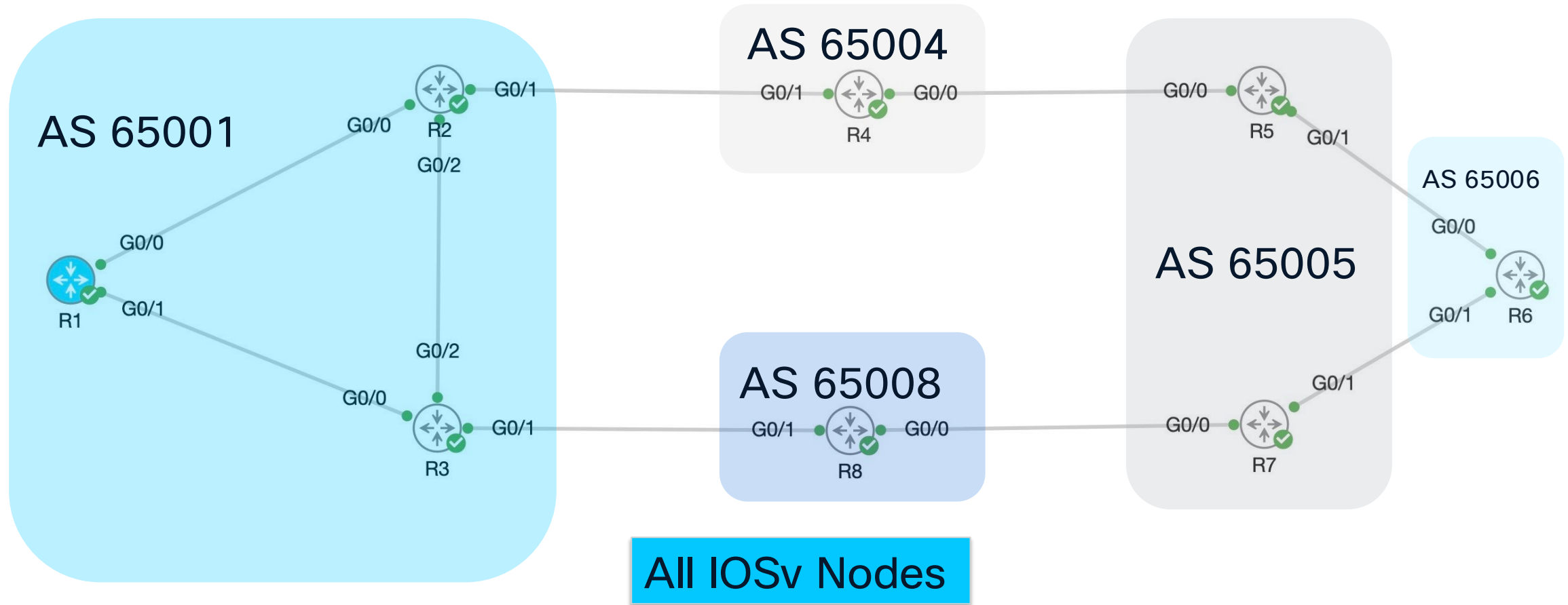
Troubleshooting BGP

- Full-mesh IBGP peering in AS 65001 uses Loopback1 interfaces instead of directly connected interfaces. R2 and R3 advertise correct next hop information to their IBGP peers. EBGP peering between R2 and R4 uses Loopback1 interfaces instead of directly connected interfaces.
- R6 should prefer the path through R7 to reach the 10.1.1.0/24 network.
- Why isn't BGP working throughout? There are five neighbor relationship issues, one routing issue, and one path selection issue.
- Find and fix the problems and make sure all loopbacks are pingable!

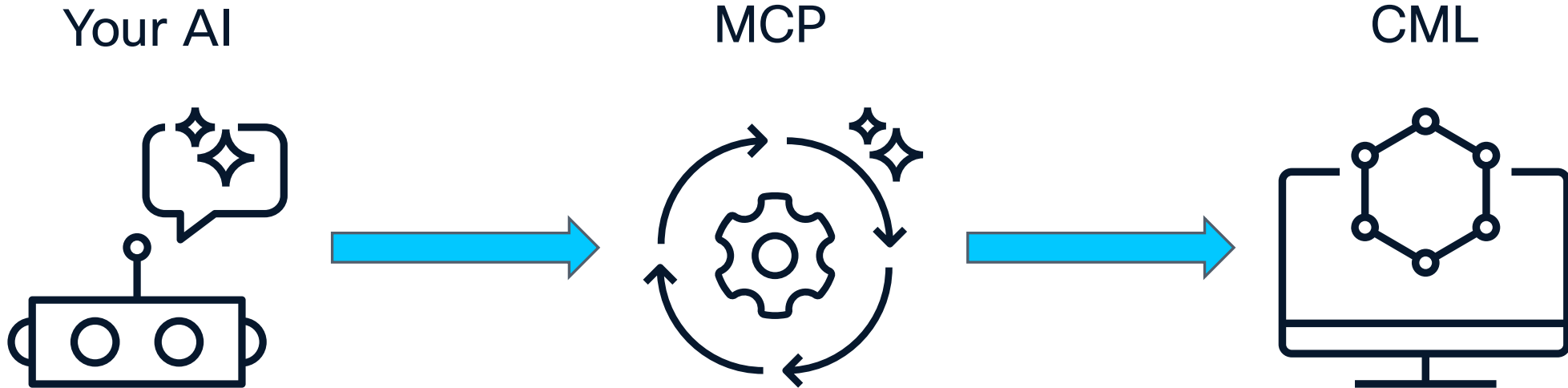


Sample Lab 2: Enterprise Advanced Routing and Switching (ENARSI)

Topology



Need More Labs? Try MCP!



CML MCP config

```
{
  "mcpServers": {
    "cml-mcp": {
      "type": "stdio",
      "command": "uvx",
      "args": [
        "cml-mcp[pyats]"
      ],
      "env": {
        "CML_URL": "https://198.18.128.11/",
        "CML_USERNAME": "admin",
        "CML_PASSWORD": "C1sco12345"
      }
    }
  }
}
```

Practice Labs On Demand, In Your Language

Cisco Modeling Labs

Workbench

VLAN PRACTICE LAB ▼

OWNER

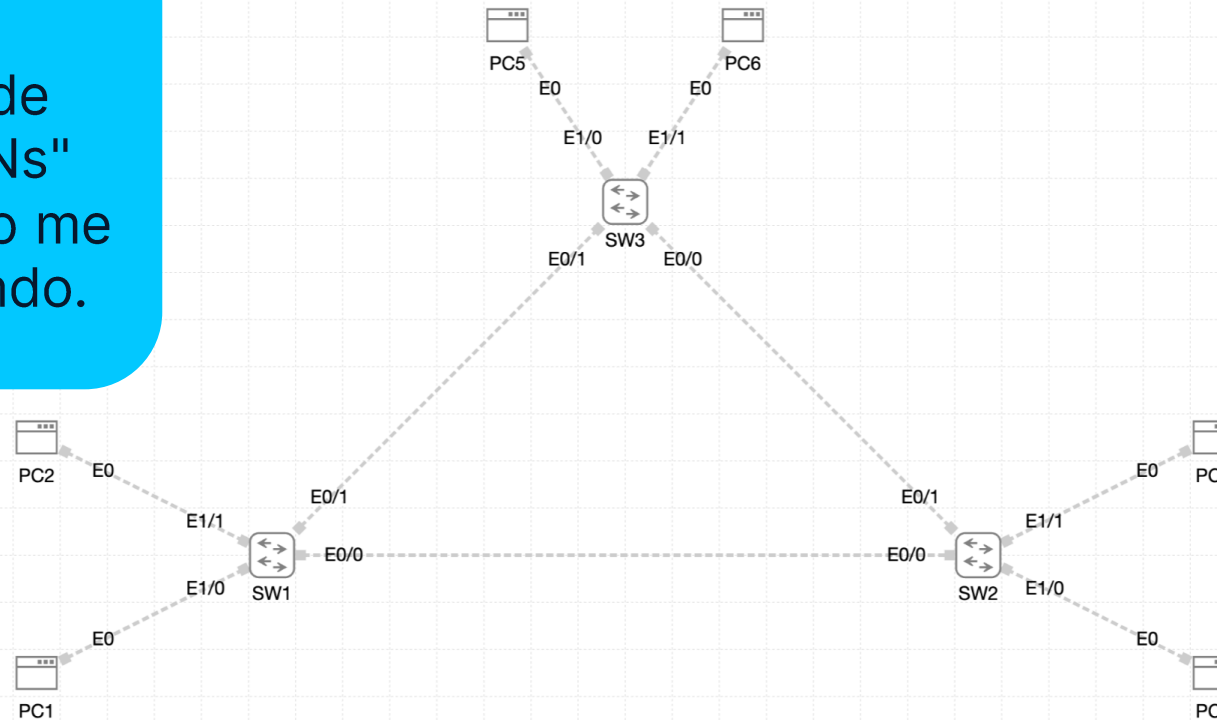
DASHBOARD

TOOLS

ADMIN

LAB NODES PANES GUIDE

Crea en CML un laboratorio CCNA de "Troubleshoot VLANs" con nodos IOL-L2. No me digas qué está fallando.



Demo – CML MCP Server

Which Outcome Do You Prefer?



AI Generated

Which Outcome Do You Prefer?



A)



Which Outcome Do You Prefer?



B)

Credly

by Pearson

Search by badge name, organization, skill, occupation or learning pathway

F

This badge was issued to [Francois Caen](#) | [View celebrations](#)

Date issued: May 01, 2026 | Expires: May 01, 2029

Share

Cisco Certified Network Professional Automation (CCNP Automation)

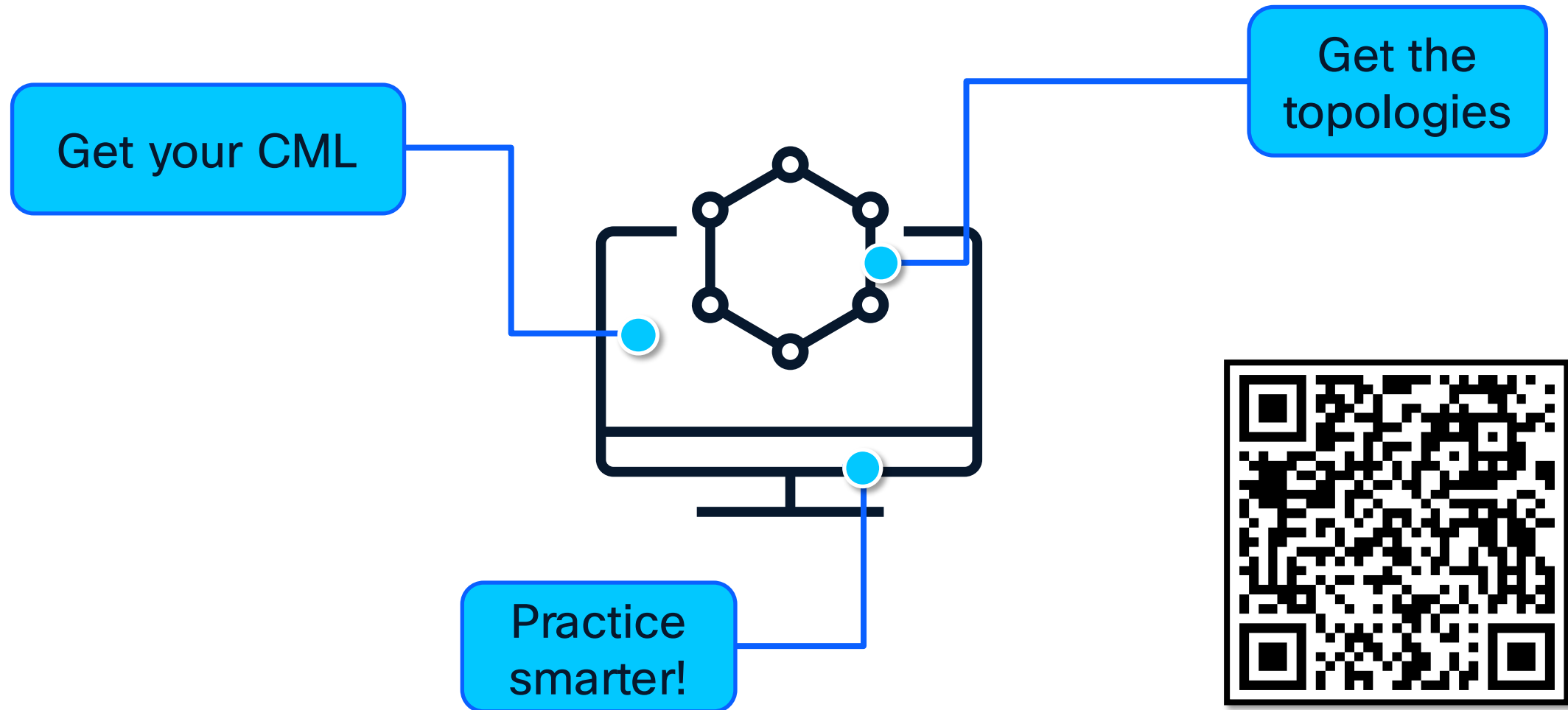
Issued by [Cisco](#)

Holders of the CCNP Automation certification have demonstrated specialized knowledge regarding the design and development of applications using Cisco APIs and platforms, and management and deployment of applications on Cisco infrastructure.

Practice Smarter!



Lab Up For Success





Compártanos su opinión.

Escanea este código para
responder la encuesta.

CISCO Live !

Thank you



